

PetroChina · KUNLUN Lubricant | TRANSFORMER OIL

KUNLUN Petro 45U

Naphthenic insulating oil — Uninhibited (antioxidant-free)

PRODUCT DATA SHEET

Product description

Petro 45U Transformer Oil is a low-viscosity, low-pour-point, antioxidant-free naphthenic base oil offering rapid heat transfer, good electrical properties and reliable oxidation stability. It is free of any synthetic additive according to the Class-U index requirements of IEC 60296:2012. The product is mainly exported and is produced under a strict quality control procedure.

Quality grade and standards

Quality grade: Standard grade.

Quality indexes according to IEC 60296:2012. Meets the Class U-30 °C specification.

Application range

Suitable for distribution transformers, generation transformers, power transformers and reactors with a Lowest Cold Start Energizing Temperature (LCSET) of -30 °C, and oil-filled electrical equipment with similar requirements, where antioxidant-free transformer oil is required. Passed assessment by DOBLE and General Electric (GE), US.

Properties & characteristics

- Relatively low viscosity and viscosity index for effective cooling and heat transfer
- Excellent heat and oxidation stability to prolong equipment life and reduce maintenance cost
- Excellent electrical insulation: high breakdown voltage and low dielectric loss factor
- Good low-temperature properties; free of pour-point depressant; pour point below -40 °C
- Free of synthetic additives per IEC 60296:2012 Class U; free of DBDS
- Passes harsh sulphur-corrosion tests (ASTM D1275B, IEC 62535, DIN 51353); free of PCB

Typical properties

Quality indexes and typical data. Quality indexes according to IEC 60296:2012. Meets the Class U-30 °C specification.

Property	Unit	Method	Min.	Max.	Typical data
Function characteristics					
Pour point	°C	ISO 3016		-40	-42
Kinematic viscosity (40 °C)	mm ² /s	ISO 3104		12	8.9
Kinematic viscosity (-30 °C)	mm ² /s	ISO 3104		1800	1000
Water content	mg/kg	IEC 60814		30	25
Breakdown voltage – untreated	kV	IEC 60156	30		40–60
Breakdown voltage – treated	kV	IEC 60156	70		>70
Density (20 °C)	kg/m ³	ISO 12185		895	876
Dielectric loss factor (90 °C)		IEC 60247		0.005	<0.001
Refining / stability characteristics					
Appearance		Visual		Clear & transparent	Clear & transparent
Acid number	mgKO H/g	IEC 62021-1		0.01	<0.01

Property	Unit	Method	Min.	Max.	Typical data
Interfacial tension	mN/m	ASTM D971	40		45
Corrosive sulphur		DIN 51353		Not corrosive	Not corrosive
Latent corrosive sulphur		IEC 62535		Not corrosive	Not corrosive
Corrosive sulphur		ASTM D1275B		Not corrosive	Not corrosive
DBDS content	mg/kg	IEC 62697-1		Not detectable	Not detectable
Antioxidant content	%	IEC 60666		Not detectable	Not detectable
2-Furfural & related compounds	mg/kg	IEC 61198		0.05	<0.05
Operation characteristics – Oxidation stability (120 °C, 164 h)					
Total acid number	mgKO H/g	IEC 61125C		1.2	0.9
Oil sludge	%	IEC 61125C		0.8	0.2
Dielectric loss factor (90 °C)		IEC 60247		0.500	0.04
Health, safety and environment					
Flash point (closed)	°C	ISO 2719	135		142
PCA content	%	IP 346		3	<3
PCB content	mg/kg	IEC 61619		Not detectable	Not detectable

Note: Typical data correspond to current production and are provided for reference; they are not a specification limit. The product is produced under a strict quality control procedure with constant product resources. Transformer oil is vulnerable to moisture and contamination – keep dry and clean, store in dedicated containers indoors or in a controlled climate, and keep free of acetylene gas.